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What is biodiesel?

What is biodiesel?

Biodiesel is a renewable fuel produced from vegetable oils or animal fats. It is a non-petroleum based diesel fuel, made up of methyl or ethyl esters of fatty acids found in vegetable oils. Biodiesel has become a standardised term referring exclusively to mono alkyl esters. There are other diesel-grade fuels of biological origin, but they are biofuels and not included in the term biodiesel [1].

The diesel engine

The diesel engine was first made by Dr Rudolph Diesel [2, 3]. He was trying to make an engine that was more efficient than the coal-burning steam-engines that were used at the time. He filed a patent in 1892, which was granted in 1893 in Germany (patent no. 677207). His first prototype, built in 1893, exploded. In 1897 his third effort was successful, and in 1900 the Otto Company of France demonstrated a Rudolph Diesel engine running on peanut oil at the World Fair in Paris. The engine was theoretically 65% more efficient than the coal engines, but in reality probably between 30–50% more efficient. By 1898 Dr Rudolph Diesel was a millionaire from franchise fees.

In the 1920s diesel engine manufacturers altered their engines to use the lower viscosity of petro-diesel (a fossil fuel) rather than vegetable oil (a biomass fuel) [4]. Petro-diesel is cheaper to produce than biofuels, so the biofuel industry waned. The side effects of burning fossil fuels are now recognised as an environmental concern, as is the realisation that fossil fuels are a finite commodity.

Properties of biodiesel

Biodiesel varies in colour from golden to dark brown. It has a high boiling point and low vapour pressure. The flash point of biodiesel is 130 °C, which is significantly higher than that of petro-diesel (64 °C) or gasoline (-45 °C). The higher flash point means that more energy is needed to make it burn. Biodiesel gels at low temperatures.

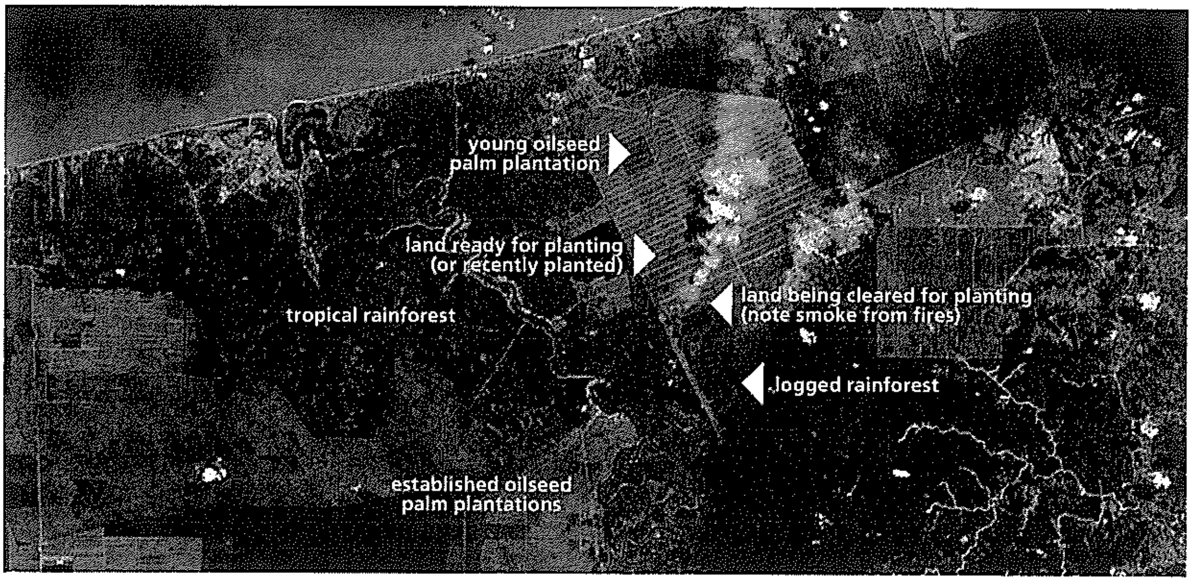
Key advantages of biodiesel

- Biodiesel can be used in normal diesel engines without modification of the engine, although new diesel engines have design modifications to maximise the biodiesel properties.
- Biodiesel is actively used as a supplement to petro-diesel to improve its lubricating properties. Petro-diesel has been modified to remove sulphur and so lost its lubricating properties. Just a 2% addition of biodiesel to petro-diesel increases the lubricity up to 50%.
- Biodiesel has been approved for up to 20% mix with petro-diesel as a world standard. However, any percentage mix up to 100% can be used, particularly with the new, modified engines.
- Biodiesel has virtually no sulphur, reduces hydrocarbon emissions, carbon monoxide, NO_x and ozone formation.
- Biodiesel reduces the particulate discharge from engines at any concentration. The particulates from petro-diesel cause lung problems such as asthma.
- Biodiesel is a renewable fuel that can be derived from any source of fatty acids, including animal fats, using a refining and production process. The final fuel produced must meet the American society of testing materials standards to be called a biodiesel.
- Biodiesel can be made from domestically produced, renewable oilseed crops such as soybeans, canola, cotton seed, mustard seed and palm.
- Some waste products, such as chicken fat, can be recycled and processed into biodiesel.
- When added as a mixture to petro-diesel, biodiesel can help to dissolve engine sediments.
- Biodiesel smells like popcorn or French fries when burnt.
- As petro-fuel prices rise, the large-scale production of biodiesel becomes more economically viable.

Key disadvantages of biodiesel

- The calorific value of biodiesel is 33 MJ/L, which is 9% lower than petro-diesel.
- Biodiesel softens and degrades rubber and elastomers with time, affecting the fuel hose and fuel pump. Engines therefore need to be modified.
- Biodiesel can dissolve some paints with time.
- Biodiesel has poor cold-flow properties. This means that the diesel engine may require some modification to warm the diesel prior to starting, or that a duel-fuel car is produced, where the petro-diesel is used initially to warm the biodiesel.

- With higher blends of biodiesel, it is possible for the fuel to get into the engine oil causing sludge formation.
- Biodiesel has a shelf life of about six months. Some biological growth can occur.
- Unfeasibly large areas of land would be required to produce sufficient biodiesel to replace fossil fuels. With current yields, an area twice the size of the United States would be required for soybean to replace America's fossil fuel usage.
- Biodiesel production requires fossil fuels at the moment.
- The switch to the production of crops for biodiesel instead of food has several consequences. It pushes the price of food crops up[5]. Biodiesel farming practices have not been assessed for environmental impact. The UN Special Report on the right to food[6] has called for a five year moratorium on biofuels labeling it a "crime against humanity" to convert food to fuel.
- Large tracts of South East Asian rainforests, particularly in Sumatra and Kalimantan, have been destroyed to make way for biofuel crops.



This aerial image shows an area in Malaysia where tropical rainforest is being converted to palm oil plantations. The rainforest is the dark green area in the upper left. Established palm oil plantations are lighter green. New plantations are brown. Note the fires that are turning rainforest debris into carbon dioxide.

Image from Central Washington Biodiesel, used with permission.

Where Australia goes from here

The Federal Government has spent $150 million on biofuel production and aims to produce 350 megalitres by 2010. Biofuels include biodiesel and ethanol-blended fuels.

BP is building a biodiesel plant in Queensland that will use tallow to produce 110 megalitres of biodiesel per annum from 2008.

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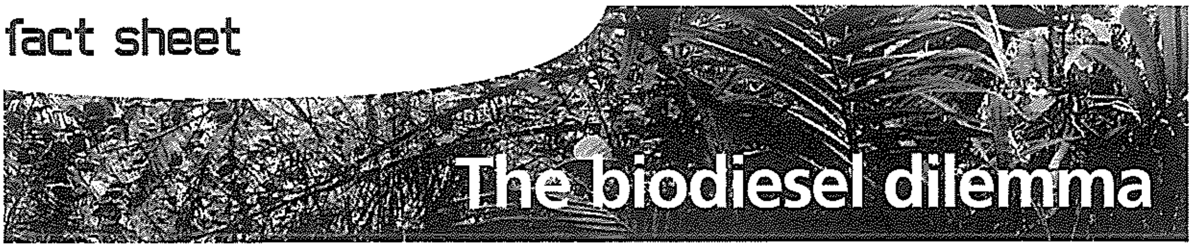


The biodiesel dilemma

The year is 2050 and the world's oil resources are diminishing at an alarming rate. As the resources become more scarce the oil price skyrockets to over US $500 per barrel.

photo: morguefile.com

You are the Minister for Sustainable Development on a group of Pacific islands that until now has been able to import enough oil to satisfy its energy needs. For over 40 years the islands have also had a small biodiesel production plant that has used the palm and coconut oil that has been produced on the islands.

A large area of the islands has always been reserved in its natural state and has been set aside as National Park. It contains several indigenous species of animal and plant that are on the *Rare and endangered species* list. It has been World Heritage listed for the last 40 years and supports a tourism industry that forms a very significant part of the islands' economy.

photo: Rarotonga Beach Bungalows, Cook Islands

Unfortunately the economy has a diminishing capacity to import its fuel needs and more and more farm land is being switched from food production to coconut and palm oil production for biodiesel. The islands are no longer self-sufficient in food production and food imports are increasing.

Members of Parliament, being pushed by lobbyists and the voting public suffering from rising food costs and thirsty for fuel, are urging you to open National Park land for agriculture. What are you going to do?

Did you know?

It was only in the 1920s that diesel engines were switched to cheaper petro-diesel. In 1900 the Otto Company of France demonstrated Dr Diesel's engine at the world exhibition in Paris using peanut oil as a fuel.

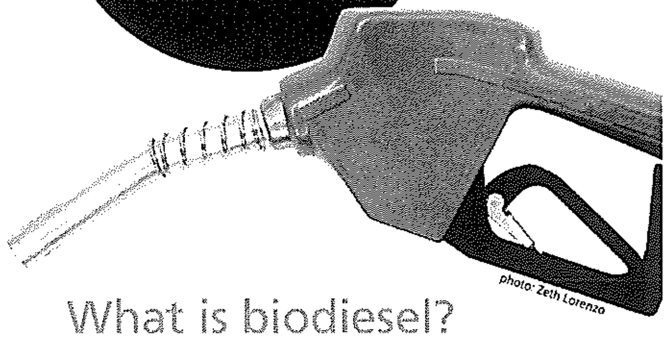


photo: Zeth Lorenzo

What is biodiesel?

Biodiesel is the name of a clean-burning alternative fuel that can be prepared from domestic or agricultural renewable resources. It can be blended with petroleum diesel to create a biodiesel blend (such as B20, B80, where the number is the percentage of biodiesel in the mix).

Biodiesel can be manufactured from vegetable oil (fresh or waste) or animal fat (tallow).

Imagine the benefits of using waste vegetable oil (WVO) from fryers and animal fat to make fuel for your car. BP is building a biodiesel plant in Queensland that will use tallow to produce 110 million litres of biodiesel per annum.

Biodiesel is biodegradable and non-toxic. It is free of sulfur and aromatic compounds that are the toxic products of burning conventional petroleum diesel.

Biodiesel can be used in conventional diesel engines without major modifications.

spice

Biodiesel emissions

Biodiesel emissions produce less pollution than conventional diesel except for nitrogen oxides (NO_2, NO, N_2O) known as NO_x. Results from a study conducted by the US Environmental Protection Agency[1] are shown below.

NB: A negative value means less of the material is produced

Quantity of polluting emissions released by burning biodiesel instead of petroleum diesel	
total unburned hydrocarbons	- 67%
carbon monoxide	- 48%
particles less than 10 microns	- 47%
NO_x	+ 10%
SO_x	- 100%
benzene compounds	- 80%
CO_2 *	- 78%

*The carbon dioxide measurement is a life cycle value. The life cycle of a fuel is a description of the processes from production through to the final use of the fuel to release energy. This includes its raw materials and emission products. Remember, it requires energy to farm the plants, extract the oil and refine the fuel before it is burnt. All of these phases produce carbon dioxide. Remember also that the plants use atmospheric carbon dioxide to make the oil. A life cycle value is important because of the role of carbon dioxide in global warming.

On the down side ... sustainability

Simply put, sustainability of an industry or other enterprise refers to whether the practice is of economic and social benefit to the local community and allows ecosystems to persist unaltered.

To produce oils such as canola, soy and palm oil on a commercial scale, large tracts of land need to be made available that are either currently being used for food production or are in their native state.

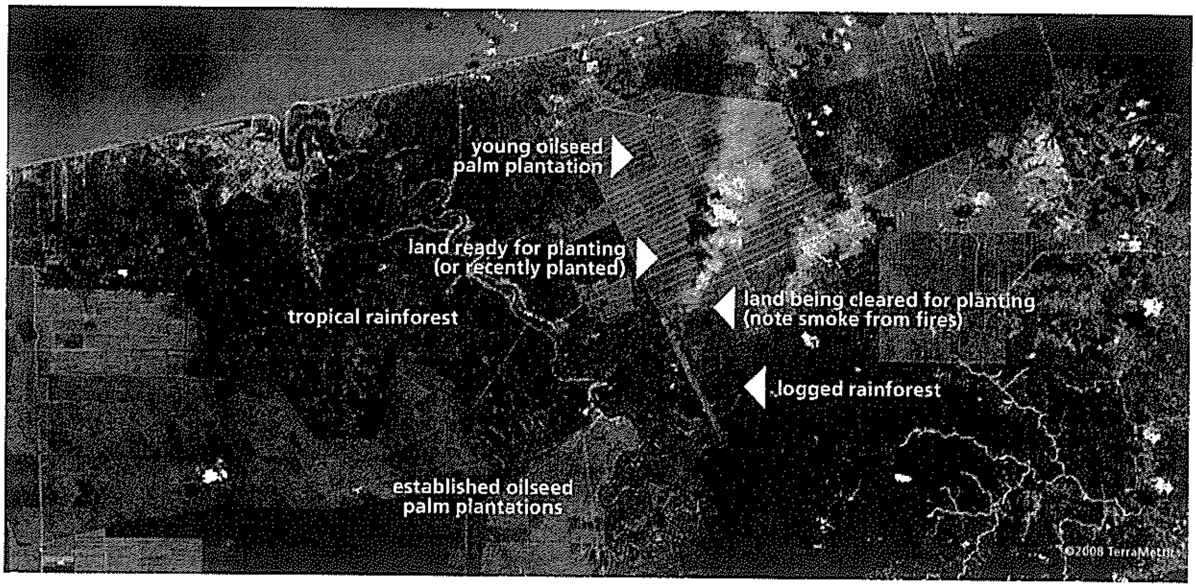
As petroleum resources diminish there will be a greater demand for biodiesel, so this is a growth industry.

So, what will the eventual outcome be?

A recent UN Special Report[3] on the right to food has called for a five year moratorium on the production of biofuels, labelling it a "crime against humanity" to convert food to fuel.

A recent OECD report[2] criticised governments for encouraging biofuel production because of the huge environmental and social costs. It has already led to the destruction of large tracts of South-East Asian rainforests. Land clearing leads to hotter, drier conditions.

At current fuel prices it is not economically sustainable to produce biodiesel on a large scale. That is, it costs more to produce than the sale price. However there are some initiatives underway to develop the system and make it environmentally and economically viable.



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Greenhouse gases

Global warming: it's a topical issue – you hear about it on the news, you see movies about it – is it really such a big deal?

Scientists are in agreement that the world is warming up: over the last hundred years the average temperature has increased by 0.75 °C. That doesn't sound like much to be so worried about, but scientists who have studied tree rings and other data think it is the hottest the world has been for at least 1300 years. There are other things that happen when the world warms up too, apart from just feeling a bit hotter. The ice caps and glaciers are melting, which has caused sea levels to rise on average 17 cm over the last century, and the sea temperature has warmed up too. This means that plants and animals have to relocate where they live or behave differently to survive. The biggest news is that scientists are predicting that the situation will get much worse, with temperatures increasing another 1.1 to 6.4 °C in the next hundred years. This will bring more melting, even higher sea levels and more extreme weather events like hurricanes, floods and droughts. This could lead to really serious outcomes like the extinction of plants and animals, islands or coastal communities being submerged and the deaths of people.

So why is this happening? Let's take a couple of steps back and look first at something called the carbon cycle.

In nature, carbon is constantly being cycled from one place to another. As you are probably aware, animals and plants produce carbon dioxide when they respire, and plants use carbon dioxide when they photosynthesise. Carbon is stored in living organisms including trees, and released when they decompose or are burnt. Carbon dioxide cycles through the ocean in the same ways.

This natural cycle is balanced, but this balance has been disturbed by human activities. Fossil fuels contain carbon and when these are burnt they release carbon dioxide, which has increased the concentration of carbon dioxide in the atmosphere. Factories, cars and electricity use are just a few factors increasing carbon dioxide levels. Over the last 250 years, since the Industrial Revolution, carbon dioxide levels have increased by more than a third. The balance of nature has been disrupted.

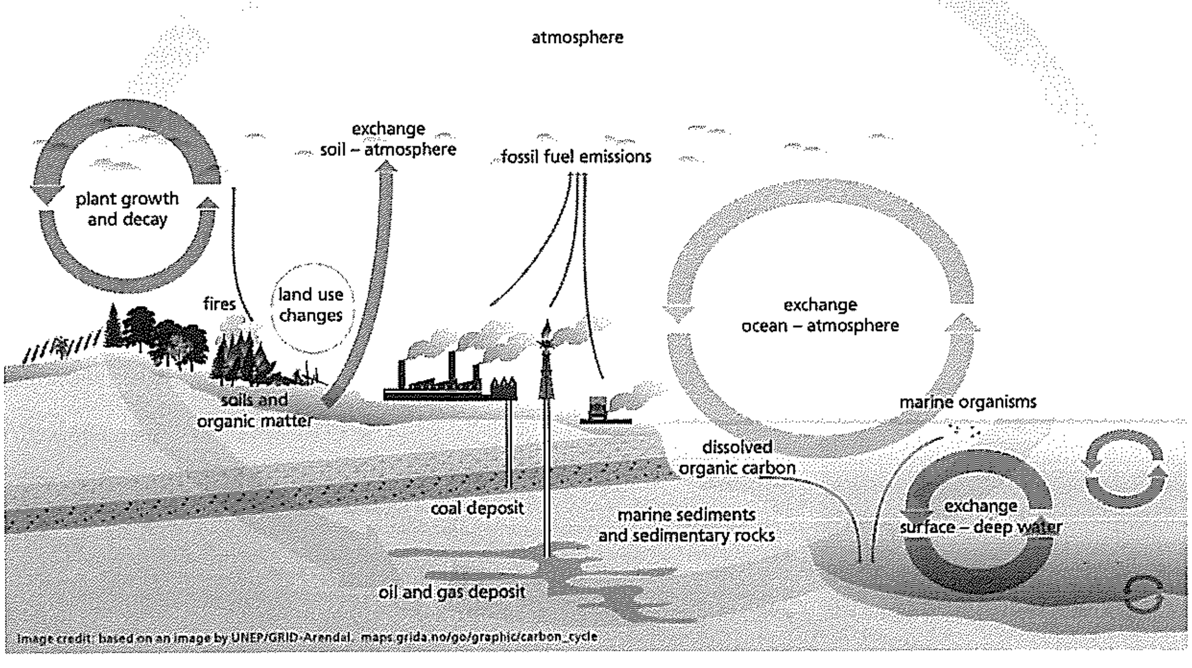
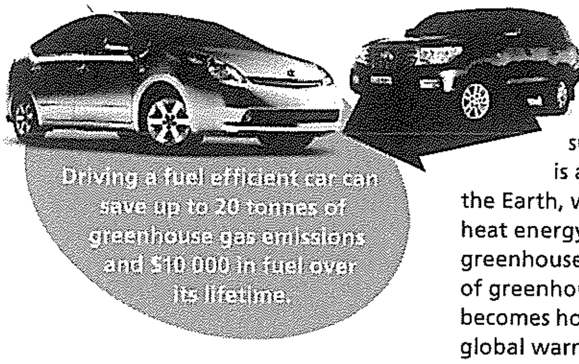
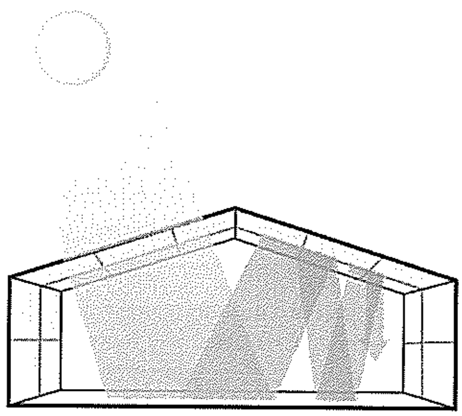
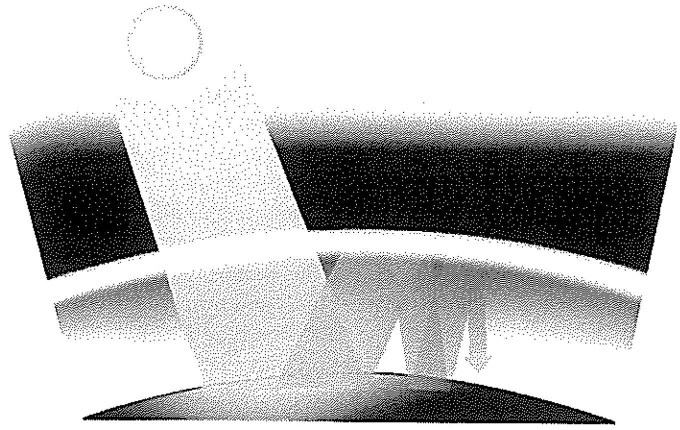


Image credit: based on an image by UNEP/GRID-Arendal, maps.grida.no/go/graphic/carbon_cycle

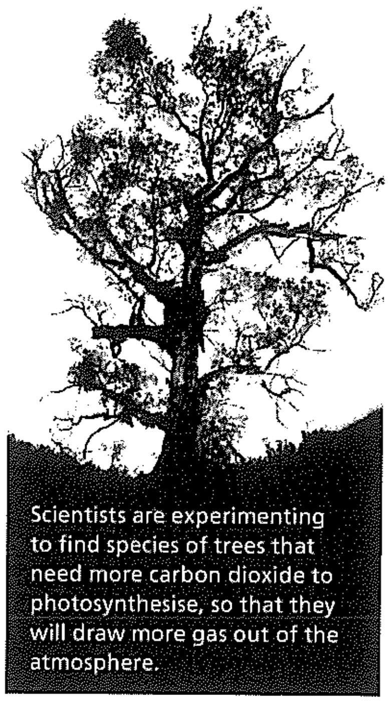


This increase of carbon dioxide is a problem because carbon dioxide is a greenhouse gas (along with methane, ozone and water vapour). The greenhouse effect is another natural phenomenon. Without it, life on earth would not exist because the planet would be too cold. The atmosphere around the earth is like a blanket. When radiation from the sun hits the atmosphere, some of it is reflected back and some is able to pass through. The radiation that passes through hits the Earth, where some of it is converted to heat energy. Much of this heat energy is radiated back into space, but some of it is absorbed by greenhouse gases and re-radiated back towards the ground. As the level of greenhouse gases increases, more radiation is trapped, and the Earth becomes hotter. This means that the more carbon dioxide there is, the more global warming there will be. Again nature has been thrown off balance.





The glass panes of a greenhouse let visible light through but trap heat energy inside. In the same way, greenhouse gases let visible light reach the Earth, but trap heat energy in the atmosphere.



So if we know that global warming is a big problem, and that it is caused by putting too many greenhouse gas emissions into the atmosphere, it should be easy to fix. We just produce less greenhouse gases. It sounds simple, but our lifestyles have become dependent on combusting large amounts of fossil fuels. Carbon dioxide is produced when electricity is made from coal, when we drive cars or ride buses and in the manufacture of all sorts of everyday items. Other greenhouse gases are produced by industry and agriculture.

As scientists and political leaders are becoming more concerned about global warming a number of possible solutions have been raised. One of these is carbon trading. The idea behind carbon trading is that producers of carbon dioxide are allocated a limit of greenhouse gas that they are allowed to make, which is called the 'cap'. If they want to produce more carbon dioxide than their cap, they must buy 'credits' off another consumer who isn't using all of theirs. This way it costs money to produce too much carbon dioxide and it saves money to be environmentally friendly. It sounds like a good concept, but there are many issues with this idea.

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How carbon trading works

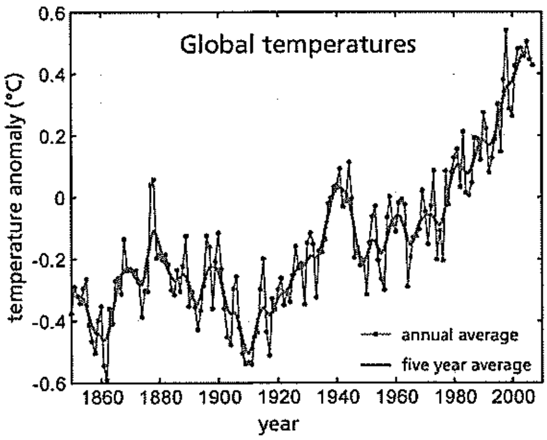
Carbon dioxide and the greenhouse effect

Global warming due to the greenhouse effect is a topical issue. Over the last few decades, the Earth's average temperature has increased significantly, possibly caused by human industry emitting increasing quantities of greenhouse gases.

The greenhouse effect is a natural phenomenon important in sustaining life. Greenhouse gases warm the Earth's surface by absorbing infrared radiation from the sun. Without this heat absorption the Earth would be too cold for animals and plants to survive. The gases include water, carbon dioxide, methane and ozone. All of these gases occur naturally, but the combustion of fossil fuels and other industrial processes have considerably increased the volume of these gases being produced, particularly carbon dioxide.

The atmosphere surrounds the Earth like a blanket. When light from the sun hits the Earth's atmosphere, some of it is reflected and some passes through to the Earth's surface. Once the radiation hits the Earth, some is absorbed and re-radiated back at a longer wavelength (infrared). A portion of this infrared radiation then travels straight back through the atmosphere, but some is absorbed by molecules of greenhouse gases, particularly water vapour and carbon dioxide. This traps the energy inside the atmosphere.

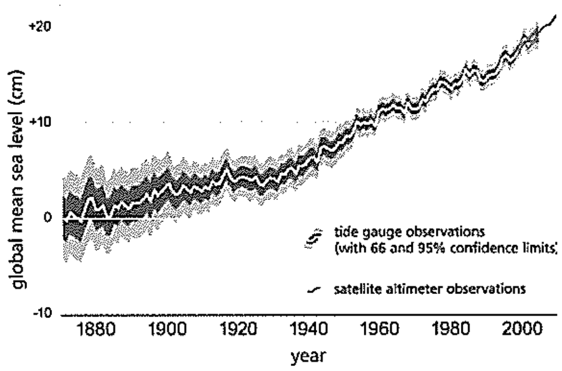
When the volume of greenhouse gases is increased, more energy is trapped inside, hence increasing the temperature of the Earth. As the Earth gets hotter, more ice melts and more water evaporates, creating more water vapour, which is also a greenhouse gas. This may lead to an accelerating effect.



Global temperature estimates for the last 150 years [2]

The average global temperature has increased by 0.75 °C over the last hundred years. It is predicted that over the next hundred years the temperature could rise by between 1.1 and 6.4 °C [1].

The predicted temperature change would cause the melting of ice caps, a rise in sea level and extreme weather events. One possible strategy to address the problem is carbon trading.



Increasing global sea levels over the 20th century [3]

Carbon trading

The basic principle behind carbon trading is to limit the total amount of greenhouse gas emissions being produced by penalising those who produce a lot of greenhouse gas and rewarding those who limit their production.

To simplify carbon trading, all greenhouse gases produced are expressed by a universal standard measurement, the carbon dioxide equivalent (CO_2 e). Various forms of emissions trading are currently being used around the world, between countries, within countries and within states. Generally an upper limit, sometimes called a cap, is set as to how much carbon dioxide a company or group may produce, and each company is given credits for that amount of carbon dioxide. Those who produce less than this amount can sell off their excess credits to those who want more, so it costs money to pollute. This encourages companies to try to cut down on their emissions and become more environmentally friendly. As the aim of carbon trading is to reduce emissions the cap is lowered over time.

The Kyoto Protocol is an international treaty signed by more than 180 countries, now including Australia but excluding the USA, agreeing to limit their production of six greenhouse gases including carbon dioxide. Each country has been set a limit of how much gas they can emit. The Kyoto Protocol allows these countries to buy emissions credits from other, usually less industrialised, countries that are producing less than their limit.

A tonne of carbon dioxide offset overseas has the same effect on reducing greenhouse gas emissions and so facilitates international carbon trading. Carbon emissions trading has been increasing at a rapid rate. According to the World Bank's Carbon Finance Unit, 374 million tonnes of carbon dioxide was traded in 2005, more than three times the amount traded in 2004 [4].

At present the only mandatory carbon trading system in the world operates between the 27 member states of the European Union. The European Union Greenhouse Gas Emission Trading Scheme commenced in 2005, and operates in accordance with the Kyoto Protocol. The scheme has established carbon dioxide caps for energy and industrial sectors, and allows carbon credits to be traded between countries.

In Australia a national carbon trading scheme is due to be implemented in 2012. The New South Wales state government independently began a Greenhouse Gas Abatement Scheme in 2003. This project requires producers of electricity to offset their carbon dioxide production. This is done by giving away energy-efficient light bulbs, managing forests and investing in other activities that reduce carbon dioxide levels.

Carbon credits for individuals

To become more aware of how we produce carbon dioxide and contribute to the greenhouse effect as individuals, it can be enlightening to calculate our 'carbon footprint'. This is the amount of carbon dioxide that we emit in a year through our daily activities. People are able to buy carbon credits to offset this amount of carbon dioxide, with each carbon credit representing one tonne of the gas. Two types of carbon credits can be purchased by individuals, either for sequestration, which usually means planting forests, or for carbon dioxide saving projects, like energy-efficient light bulbs.

Problems with carbon trading

In theory, carbon trading is a useful concept, but in practice there are a number of issues that need to be overcome before it works well. These include:

- monitoring the amount of emissions produced to ensure they equal or are less than the limit set,
- high polluters buying extra credits easily compromises a reduction in overall emission levels,
- industry passing on the extra expense of carbon trading to consumers,
- establishing an appropriate cap level – if set too high carbon credits become meaningless,
- high administrative costs,
- controlling corruption, and
- individual or corporate carbon credits risk deflecting attention from the greater issue of carbon dioxide production.

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